

#Maintaining User/Server State (cookies)

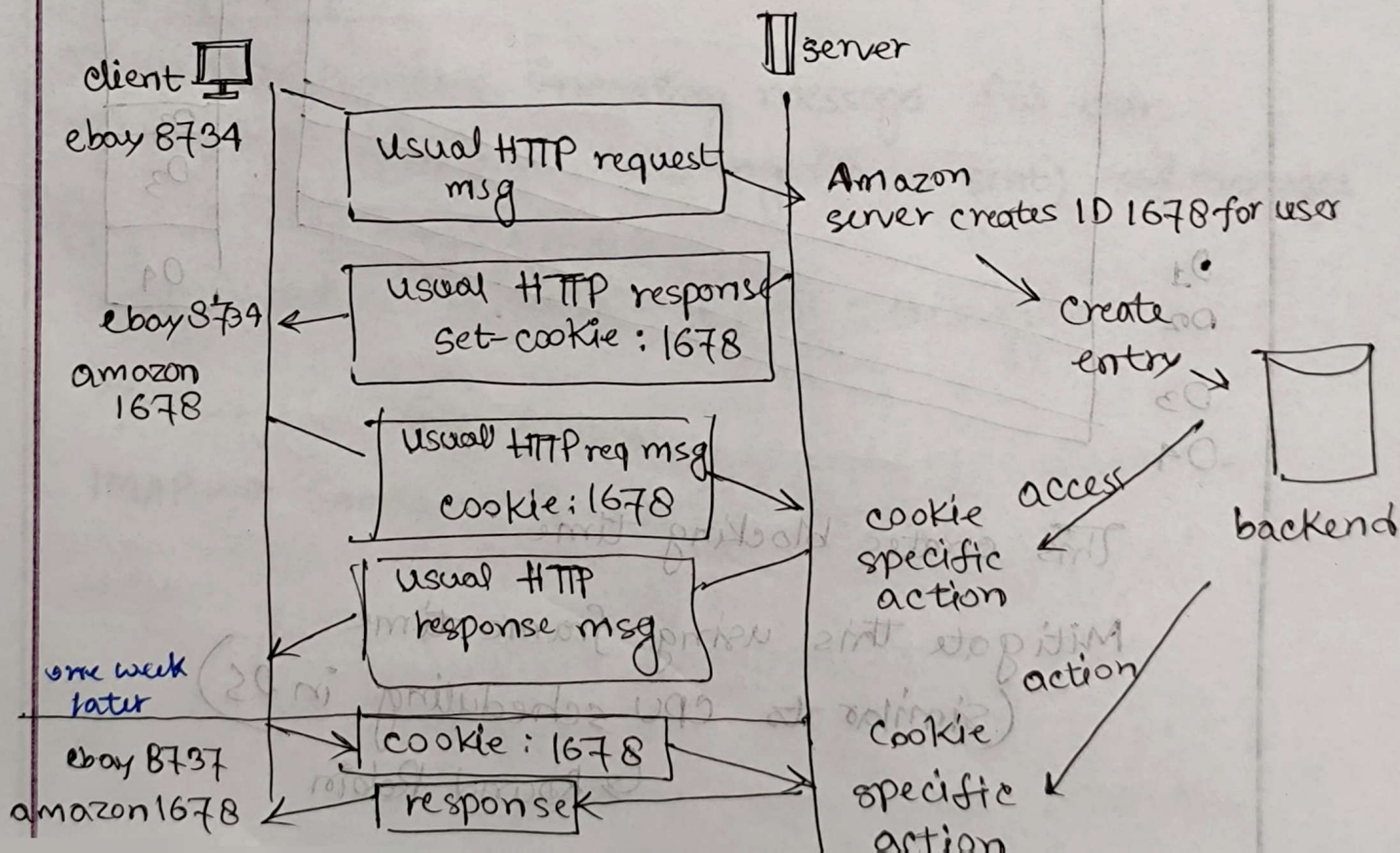
HTTP is stateless &

Websites and client browsers use **cookies** to maintain some state between transaction.

four components :

- ① Cookie header line of HTTP response message,
- ② Cookie header line in next HTTP request message,
- ③ Cookie file kept on user's host, managed by user's browser,
- ④ Back-end Database of the website.

Cookie Mechanism



Web Caches (Proxy Servers)

Caching Example

conditional GET

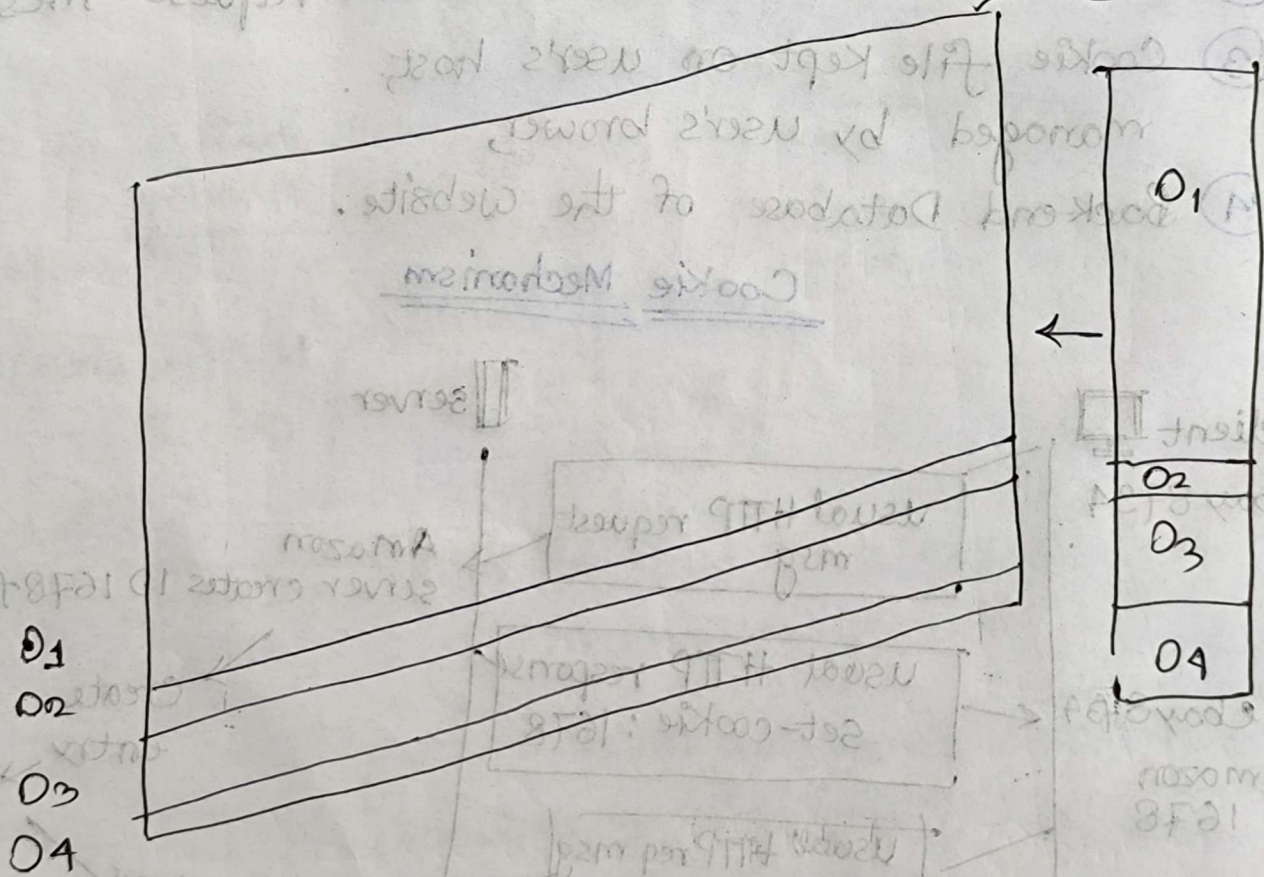
HTTP/2 : Mitigating HOL blocking

client



Get 04 Get 03 Get 02 Get 01

Server



This creates blocking time

Mitigate this using frame time

(similar to CPU scheduling in OS)

↳ Round Robin

(SMTP) E-mail

③ Major Components

{ Routing vs. Forwarding }

- ① User Agent
- ② Mail Servers
- ③ SMTP (Simple Mail Transfer Protocol)

#User Agent

- ① Mail Reader
- ② Composing, Editing, Reading mail messages
- ③ Outgoing/incoming messages stored on server.

#Mail Servers

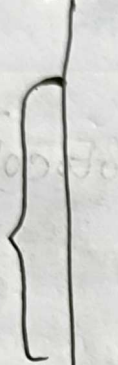
- ① mailbox contains incoming messages for user
- ② message queue of outgoing (to be sent) mail messages
- ③ SMTP protocol between mail servers to send email-messages.

POP → Post Office Protocol

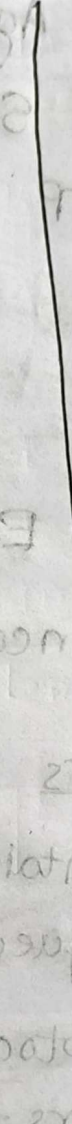
IMAP → Something Something Protocol

#SMTP RFC (5321)

"client"
SMTP server



"server"
SMTP server



checking protocols using TELNET

→ google

① Syntactical Mistake

② Semantic Mistake

#SAME SMTP Interaction

#SMTP status codes

#Retrieving Email: Mail Access Protocol

#DNS (Domain Name System)

ipconfig /all

local DNS → convert hostname to IP address

IP Packet → read/forward over the network

32bit IP addresses → 2^{32} IP addresses

4Giga " "

↳ IPv4

DNS: Services & Structure

Services:

↳ Host name ↔ IP address

↳ Host aliasing

↳ Canonical; aliasname

↳ Mail server aliasing

↳ Load distribution

↳ Replicated Webservers:

Many IP addresses

Correspond to my name.